

MARY COTRUPI

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Research interests: Quantum software & algorithms, quantum HPC

EDUCATION

Rensselaer Polytechnic Institute (RPI)

B.S. in Computer Science; Minors: Quantum Computing, IT & Web Science

Troy, NY

Expected May 2027

Undergraduate GPA: 3.8/4.0 | Dean's List | RPI Founders Award of Excellence (top 1%)

Related coursework: Computing & Quantum Computing; Quantum Information Processing; Quantum Computer Organization; Distributed & Parallel Computing; Algorithms; Linear Algebra; Machine Learning from Data; Discrete Mathematics & Probability

The Chinese University of Hong Kong (CUHK) – Computer Science Study Abroad, Hong Kong

Spring 2026

TECHNICAL SKILLS

Programming & scientific computing: Python, C++, CUDA, CMake, NumPy, SciPy, JAX, Git/GitHub, Linux

HPC & performance: Slurm, MPI, OpenMP, NVIDIA Nsight, multi-GPU computing

Quantum computing: Qiskit, Pennylane, Qualtran, Stim, Aer, PyMatching, IBM Quantum, resource estimation

RESEARCH EXPERIENCE

Future of Computing Summer Intern (FCSI) | National Security Agency

Summer 2026

Towards Efficient Low-Weight Decoding: A Blocked Fast Walsh-Hadamard Transform

- Co-developed a novel blocked fast Walsh-Hadamard transform, an exact set-covering decoder for solving the low-weight binary Closest Vector Problem; at $n = 64$, $m \sim 2,000,000$, $t = 8$, reduced estimated operations by approx. $730\times$ versus optimized Gray-code enumeration. Benchmarked CUDA/GPU implementations on A100 clusters, achieving a $\sim 44\times$ speedup at $n = 48$, $t = 8$ while evaluating covering designs, pruning, and performance tradeoffs.
- Built an end-to-end Decoded Quantum Interferometry quantum algorithm codebase with our novel decoder (**BFWHT-DQI**), benchmarking against brute-force and full-FWHT. Cut estimated logical qubits by $\sim 700\times$ vs. full FWHT (64×64 , $t = 8$), testing on IBM's 156-qubit r2 Heron QPU.

Quantum Machine Learning Research Intern | RPI School of Science

Summer 2025

Quantum machine learning kernels for network intrusion detection: a comparative NISQ application

- Designed a Qiskit quantum-kernel/SVM pipeline for network intrusion detection; achieved 85% average accuracy on RPI's 127-qubit IBM Quantum System One, within 6% of noiseless simulation, benchmarked against classical RBF SVMs.
- Built the reproducible **QML-NIDS** research codebase for data preparation, hardware execution, benchmarking, and analysis; first-authored and presented the resulting AAAI QIML proceedings paper.

Quantum Computing Researcher (Quantum Chemistry)

Aug 2026 – Present

- Quantum chemistry simulation research with Dr. Osama Raisuddin—investigating dynamic quantum circuits to reduce circuit depth; implemented a Pauli-exponential decomposition in Qiskit for molecular energy estimation in VQEs and optimized for performance on real quantum hardware.

Cybersecurity Research Assistant | Rensselaer Cybersecurity Collaboratory

Aug 2024 – May 2025

- Developed augmented-reality CTF challenges in steganography, DNS poisoning, and MFA bypass; coauthored a 2026 CISSE article and ASIA proceedings on generative-AI use and cybersecurity pedagogy, presenting related work at BSides St. Pete.

SELECTED PUBLICATIONS & PRESENTATIONS

- Cotrupi, M.**, Aurilio, A., Sanpietro, S., Torres, F., Kasturi, A., Colognato, Q., & Callahan, B. R. (forthcoming). *Hardware Robustness of Quantum Machine Learning Kernels for Malware Detection*. *Proceedings of the 2026 AAAI Fall Symposium Series (QIML)*. Accepted; Arlington, VA, Nov. 5–7, 2026.
- Cotrupi, M. L.**, & Callahan, B. R. (2025). *Towards Practical Quantum Kernels for Network Intrusion Detection*. *Proceedings of the AAAI Symposium Series*, 7(1), 339–342. doi:10.1609/aaaiss.v7i1.36903 (3 citations)
- Sugerman, S., Joseph, S., Colognato, Q., **Cotrupi, M. L.**, et al. (2026). *A Case Study for Combating Student Overuse of Generative Artificial Intelligence in Cybersecurity Educational Activities Using Augmented Reality Capture-the-Flag Development*. *Journal of the Colloquium for Information Systems Security Education*, 13(1), Article 11. doi:10.53735/cisse.v13i1.216
- Cotrupi, M. L.**, Gutti, M., Ralph, B., & Sandberg, K. (2026). *Towards Efficient Low-Weight Decoding: A Blocked Fast Walsh-Hadamard Transform*. FCSI 2026 final technical report.
- ISC2 Security Congress 2026** (upcoming) – *A Quantum Cybersecurity Platform to Protect Organizations That You Can Use Today*, with B. R. Callahan; Aurora, CO, Oct. 26, 2026.

LEADERSHIP, ACTIVITIES & HONORS

- Cyber Sentinel Challenge**, U.S. Department of Defense – Top 5% among 2,100+ selected competitors (2025).
- Admin & Student Leader**, Rensselaer Cybersecurity Collaboratory – Led technical CTF sessions in reverse engineering, insider threats, and cyber forensics (2023–2025).
- Principal Second Violinist**, Rensselaer Orchestra – Section leader/assistant leader, ~ 15 violinists (2023–Present).
- Honors & professional societies** – RPI Upsilon Pi Epsilon (CS), RPI Gamma Nu Eta (ITWS), ISC2 Capital District NY.